STUDYLAB HUB · AP & COLLEGE BIOLOGY · UNIT 1 (CHAPTERS 1–5)

Unit 1: Chemistry of Life

Scientific Method · Water & Hydrogen Bonding · Carbon · Carbohydrates, Lipids, Proteins & Nucleic Acids

AP & College Biology — Unit 1 (Chapters 1–5): Chemistry of Life

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1. Unit 1 Roadmap (Chapters 1–5)

Chapter / CED Topic	Core Concept	What Exams Actually Test
Ch. 1 · The Study of Life	Scientific method, hypotheses, null vs. alternative hypotheses, experimental controls	Identifying independent/dependent variables, positive vs. negative controls, and interpreting error bars ($\pm 2 \text{ SE}_x$)
Ch. 2 · Water & Life (CED 1.1)	Polar covalent O–H bonds, intermolecular hydrogen bonding, emergent properties of water	Predicting what happens when polarity or H-bonding is disrupted (cohesion, adhesion, surface tension, high specific heat, evaporative cooling, ice density, pH & buffers)
Ch. 3–4 · Carbon & Functional Groups (CED 1.2)	Tetravalent carbon (4 valence e^-), functional groups, dehydration synthesis vs. hydrolysis	Recognizing hydroxyl, carbonyl, carboxyl (acidic), amino (basic), sulfhydryl (disulfide bridges), phosphate (negative charge/ATP), and methyl (nonpolar/epigenetic) groups
Ch. 5 · Biological Macromolecules (CED 1.3–1.6)	Carbohydrates, Lipids, Proteins (1°–4° structure), Nucleic Acids (DNA vs. RNA)	Directionality ($5' \to 3'$ vs. N-terminus $\to$ C-terminus), monomer-to-polymer bond types, and how R-group mutations alter protein folding

2. Chapter 1: Experimental Design & Statistical Reasoning

1. **Null Hypothesis (H_0) vs. Alternative Hypothesis (H_a)**:

- **H_0**: States there is **no statistically significant difference** between groups (any observed difference is due to random chance).
- **H_a**: States that the independent variable causes a real, measurable effect on the dependent variable.

1. **Controls on Free-Response Questions (FRQs)**:

- **Negative Control**: Group receiving no treatment (or placebo/solvent only) to establish the baseline rate and rule out confounding variables.
- **Positive Control**: Group receiving a treatment known to produce the expected effect, proving the assay/reagent actually works.

1. **Standard Error of the Mean ($\text{SE}_{\bar{x}}$)**:

- When error bars ($\pm 2\,\text{SE}_{\bar{x}}$, roughly 95% confidence intervals) between two treatment groups **do not overlap**, the difference between means is likely statistically significant.

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3. Chapter 2 (CED 1.1): Structure of Water & Hydrogen Bonding

- **Inside one water molecule**: Oxygen is much more electronegative than hydrogen ($\chi_{\text{O}} \approx 3.44$ vs. $\chi_{\text{H}} \approx 2.20$). Electrons spend more time near oxygen, creating a **polar covalent bond** with partial negative charge (δ^-) on O and partial positive charges (δ^+) on each H.
- **Between water molecules**: The δ^+ H of one molecule attracts the δ^- lone pairs on O of a neighboring molecule, forming a **hydrogen bond** (intermolecular attraction, ~5–10% as strong as a covalent bond).

Property of Water	Molecular Mechanism	Biological Consequence
Cohesion	H-bonds hold water molecules to **each other**	Creates high surface tension; pulls water column upward during transpiration in xylem
Adhesion	H-bonds form between water and **other polar/charged surfaces** (e.g., cellulose cell walls)	Resists downward pull of gravity in plant xylem (capillary action)
High Specific Heat	Heat energy must first break many H-bonds before increasing kinetic energy (temperature)	Stabilizes ocean and intracellular temperatures
High Heat of Vaporization	Breaking H-bonds for liquid $\to$ gas requires large energy input	Evaporative cooling (sweating, leaf transpiration) removes excess heat
Ice Floats (Solid Less Dense)	At 0°C, each water molecule locks into a crystalline tetrahedral lattice with 4 neighbors	Ice forms an insulating top layer on lakes so aquatic life survives below
Versatile Polar Solvent	Hydration shells surround ions (Na^+, Cl^-) and polar molecules	Dissolves blood plasma solutes and cytosolic reactants (hydrophilic vs. hydrophobic)

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4. Chapters 3–5 (CED 1.2–1.6): Dehydration Synthesis, Hydrolysis & The 4 Macromolecules

- **Dehydration Synthesis (Condensation)**: Joins monomers into a polymer by **removing one H_2O molecule** per covalent bond formed (requires energy + enzymes).
- **Hydrolysis**: Breaks a polymer back into monomers by **adding one H_2O molecule** across the covalent bond (releases free energy; digestion).

Macromolecule	Elements	Monomer / Subunit	Covalent Linkage	Key Examples & Directionality
Carbohydrates	C, H, O ($1:2:1$)	Monosaccharides (glucose, fructose, ribose)	**Glycosidic linkage** (α-1,4 in starch/glycogen vs. β-1,4 in cellulose)	Starch & glycogen (energy storage); cellulose & chitin (structural support; humans lack cellulase to hydrolyze β-1,4 linkages)
Lipids *(not true polymers)*	C, H, O (high H:O ratio; phospholipids add P, N)	Glycerol + fatty acids (or 4 fused carbon rings in steroids)	**Ester linkage**	Saturated (no C=C double bonds, straight chains, solid at room temp) vs. Unsaturated (*cis* C=C double bonds create kinks, liquid/fluid membranes); **Phospholipids** are amphipathic (hydrophilic head + hydrophobic tails)
Proteins	C, H, O, N, S	Amino acids (central α-C, amino group, carboxyl group, R-group)	**Peptide bond** (C-terminus carboxyl to N-terminus amino)	Synthesized **N-terminus ($-\text{NH}_3^+$) $\to$ C-terminus ($-\text{COO}^-$)**; R-groups classified as nonpolar/hydrophobic, polar/hydrophilic, or charged (acidic/basic)
Nucleic Acids	C, H, O, N, P	Nucleotides (5-C pentose sugar + N-base + phosphate group)	**Phosphodiester linkage** ($3'\text{-OH}$ of one sugar to $5'\text{-phosphate}$ of next)	Synthesized **$5' \to 3'$**; antiparallel DNA strands held by H-bonds (**A=T** has 2 H-bonds; **G$\equiv$C** has 3 H-bonds, so higher GC% = higher melting temp T_m)

5. Protein Folding Hierarchy ($1^\circ \to 4^\circ$) & Mutation Trace

1. **Primary (1°) Structure**: Exact linear sequence of amino acids linked by covalent **peptide bonds** (determined directly by mRNA codons).
2. **Secondary (2°) Structure**: Local α-helices and β-pleated sheets stabilized by **hydrogen bonds along the polypeptide backbone** (between carbonyl O and amide H — *not* R-groups).
3. **Tertiary (3°) Structure**: Overall 3D fold of a single polypeptide driven by **R-group (side-chain) interactions**: hydrophobic clustering inside the core, H-bonds, ionic salt bridges, van der Waals forces, and covalent **disulfide bridges ($-\text{S}-\text{S}-$)** between cysteine residues.
4. **Quaternary (4°) Structure**: Assembly of **two or more polypeptide subunits** into one functional complex (e.g., hemoglobin's $2\alpha + 2\beta$ subunits).

> **Classic AP FRQ Mutation Rule**:
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> - Replacing a hydrophobic amino acid (e.g., Valine) with another similarly sized hydrophobic amino acid (e.g., Leucine) often preserves 3° folding (silent/conservative effect).

> - Replacing a charged/polar amino acid (e.g., Glutamate, negatively charged) with a nonpolar hydrophobic amino acid (e.g., Valine, as in sickle-cell hemoglobin) flips the local polarity, alters $3^\circ / 4^\circ$ folding, and disrupts active-site or binding geometry.

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6. Worked Exam Practice (MCQs + Scored FRQ)

Practice Multiple-Choice

1. **A DNA molecule from Organism A contains 18% Adenine (A). What percentage of the bases in this double-stranded DNA molecule are Cytosine (C), and how would its thermal stability compare to a DNA molecule with 35% Adenine?**

- **Answer**: By Chargaff's rules, $\%\text{A} = \%\text{T} = 18\%$ (36% total), leaving 64% for $\text{G} + \text{C}$, so **$\%\text{C} = 32\%$**. Because $\text{G}\equiv\text{C}$ pairs form **3 hydrogen bonds** (vs. 2 for $\text{A}=\text{T}$), Organism A's DNA (64% GC) has **higher thermal stability** (higher denaturation temperature) than the 35% A DNA (30% GC).

1. **An enzyme resides inside the hydrophobic interior of the plasma membrane. Which amino acid substitution in its exterior transmembrane domain is most likely to disrupt its placement in the membrane?**

- **Answer**: Substituting a **nonpolar/hydrophobic** amino acid (e.g., Phenylalanine or Isoleucine) with a **charged/hydrophilic** amino acid (e.g., Lysine or Aspartate), which is energetically unfavorable inside the nonpolar fatty-acid tail core.

Worked Free-Response Question (FRQ)

Prompt: Researchers measure the activity of a bacterial hydrolytic enzyme across temperatures from 10°C to 80°C. Activity peaks at 37°C and drops to zero at 70°C.

- **(a) Identify the specific covalent bond broken during the reaction catalyzed by a protease that hydrolyzes a polypeptide, and state whether water is consumed or produced.**
- **(b) Explain at the molecular level why heating the enzyme to 70°C reduces activity to zero even though its primary (1°) structure remains intact.**

Scoring Rubric:

- **(a)** Breaks **peptide bonds** between amino acids; **water is consumed** (added across the bond during hydrolysis).
- **(b)** High thermal kinetic energy disrupts weak noncovalent interactions (**hydrogen bonds, ionic bonds, and hydrophobic interactions**) among R-groups and backbone atoms, unfolding the **2° and 3° structure (denaturation)** and distorting the active site without breaking covalent peptide bonds (1° sequence).